

SQL Execution & Internal Flow

A SQL query is a statement that requests data or instructs the database to perform an operation.

Query vs Statement

This interview question appears surprisingly often.

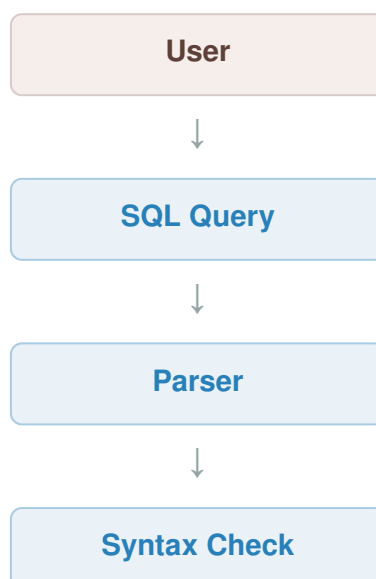
Statement

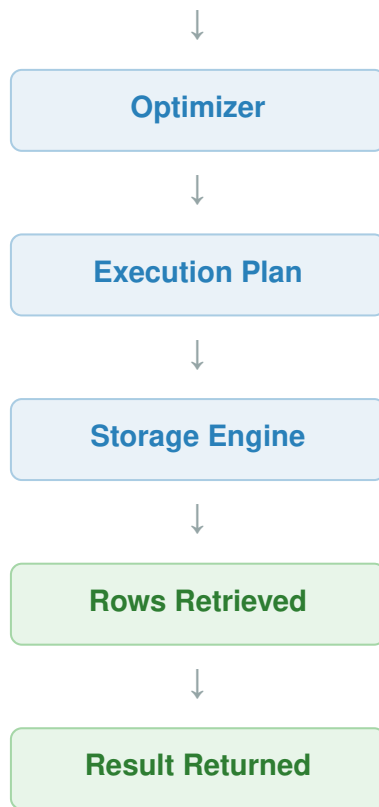
A SQL statement is any valid SQL command. A query is generally a SQL statement that asks the database for information.

```
SELECT *  
FROM Students;
```

the database does much more than simply read the table.

Internal flow:





STEP 1 — PARSER

Checks:

- SQL syntax
- keywords
- table names
- column names

STEP 2 — OPTIMIZER

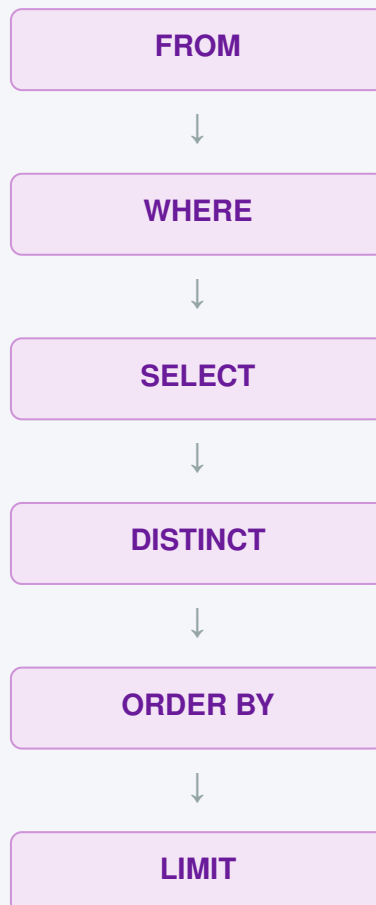
Decides:

- Which index to use
- Whether to scan the whole table
- Join order (later)

STEP 3 — STORAGE ENGINE

Actually reads data from disk or memory.

Execution Order



Why Isn't SQL Written in Execution Order?

Because SQL is a declarative language.
You describe what you want, not how to get it.

What is the WHERE Clause?

The WHERE clause filters rows.
It tells the database:

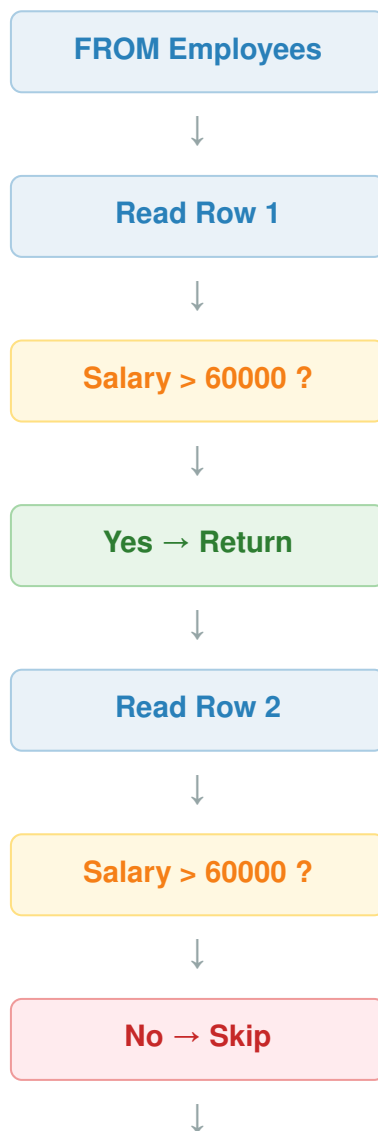
"Return only those rows that satisfy this condition."

Internal Working

Suppose:

```
SELECT *  
FROM Employees  
WHERE Salary > 60000;
```

Logical execution:



Continue until end

If no suitable index exists, this is called a table scan (or full table scan).

If an appropriate index exists, the database may jump directly to matching rows instead.

Every WHERE condition ultimately evaluates to one of three values:

- **TRUE**
- **FALSE**
- **UNKNOWN (NULL)**

Only rows with **TRUE** survive.

NOT UNKNOWN = TRUE

This is incorrect.

It remains UNKNOWN.

Wrong

```
WHERE NOT Department='ECE'  
AND Salary>50000;
```

This is interpreted as:

```
WHERE (NOT Department='ECE')  
AND Salary>50000;
```

If your intent is different, use parentheses.

Example:

```
WHERE NOT  
(  
  Department='ECE'  
  AND Salary>50000  
);
```

SQL Operator Precedence

For the operators you've learned so far, the order is:

